Analysis report
Compilatio Magister+ | Amrita Vishwa Vidyapeetham (PILOT)

Performance of natural language processing models in extracting important clinical information from a free text radiology report

ID : 63b4de89454d725adde3b56e170de361f4161530



8% Suspicious texts

File name : ASSIGN-2_24108_102.txt
Original file size : 38.37 KB
Number of words : 1,010

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Summary (section 1/3)

Location of suspect texts in the document :



Included in the suspicious text score :

Similarities — 0%

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Passages with similarities to sources found in different collections.

AI detection — 7%

Texts with stylistically similar formulations to AI-generated text.
This rate is an indicator, not proof. Check with the author that he/she has mastered the knowledge mentioned in the document.

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Passages in which some of the vocabulary used is not part of the language dictionary. This may be an attempt by the author to modify the text to make detection impossible.

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Texts between quotes — 0%

Passages between quotation marks, often revealing a quotation.

Sources of similarities (section 2/3)

Similarities

0%

Passages with similarities to sources found in different collections.



23CSE301 – Machine Learning Assignment – 2

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Roll No : **CH.SC.U4CSE24108**

Paper Id : **102**

Paper Title : **Performance of natural language processing models in extracting important clinical information from a free text radiology report**

PHASE – 2 ASSIGNMENT

Feature selection and identification of learning algorithm

My project is around extracting useful medical information from chest X ray reports using NLP. The input is the text written by the radiologist and the output is the status of 14 chest related findings. Each of those findings can be present, absent or uncertain. As there can be multiple findings in a single report this is a multi label classification problem (e.g. a report could have pneumonia as present and pleural effusion as absent). For this project I've selected the OpenL Chest X Ray collection from Indiana University. It has around 3955 radiology reports and 7470 X rays. The dataset is publicly available and patient identifiers have been stripped. The reports will have a comparison, impressions, findings and indication section. I'm going to

be using the findings and the impression sections as these contain the radiologist's observations and the final diagnosis. The project is based on the report text so I won't be using the X rays at all. The features I'm going to use are the report text wordpiece tokens, ClinicalBERT representations and some key expressions relating to negation and uncertainty. The wordpiece tokens break the words up into smaller units which the model can work with. I'm going to use a max sequence length of 128 tokens as the reports are generally quite short. ClinicalBERT will produce contextual representations of the text and the [CLS] token representation will be used for classification. Expressions such as 'no', 'without' and 'cannot exclude' are important as they can change the meaning of a finding. MESH and Radlex will be used as the medical terminology resources rather than as direct input features. The base paper reported an AUC of 0.99 for single site tasks versus 0.95 for multi site tasks. This was the reason I chose to focus on chest radiology (though it by no means guarantees that my model will achieve the same performance).

Existing Algorithms and Literature survey

NLP methods for radiology reports have evolved from rule based systems through to traditional machine learning, through to deep learning (transformers) and now to large language models. The early methods were predominantly reliant on medical dictionaries and manually written rules as there were very limited labelled clinical datasets. Subsequent work has looked at machine learning methods to detect patterns in the medical text. More recent work has looked at the use of transformer models and LLMs to understand the context of the medical language. Cotik et al 2015 looked at the classification of Spanish radiology reports using rules and medical vocab lists. They were able to extract information without needing a large training set, though their method was based on hand-coded rules and used only 248 reports. Hassanpour and Langlotz 2016 looked at information extraction from reports from multiple institutions using CRFs and MaxEnt. This work did show the value of traditional ML approaches but again required manually annotated data and carefully hand-crafted features. Chapman et al 2017 used keywords and rules to identify evidence of surgical site infection in radiology reports. The study did show that NLP could be used to support clinical tasks, though it was clear that some actual findings were being missed. Park et al 2022 used deep learning to extract lung cancer staging information from PET CT reports and reported superior performance compared to rule based methods. The study did use private data from a single institution and no external validation was performed. Hu et al 2024 looked at using ChatGPT for information extraction via zero-shot prompting. This approach did reduce the need for manually labelled training data, but there were concerns around privacy, API costs and the need for careful prompt design. Le Guellec et al 2024 evaluated an open source LLM for information extraction from radiology reports. Local deployment was advantageous in terms of control of sensitive data but the approach did require significant compute resources and again lacked external validation. My project looks at how different models perform by using a public chest radiology dataset.

Identification of learning Algorithm and technique

The base paper doesn't train a new NLP model. The authors have done a systematic review and meta-analysis of the 28 published studies (approximately 51187 reports and 421692 extracted clinical entities). They've followed the PRISMA DTA guidelines and used a bivariate random effects meta-analysis in Stata SE 18 to pool the sensitivity and specificity. The pooled results were a sensitivity of ~91%, a specificity of 96% and an AUC of 0.98. This is a very good overall result, but there was a lot of variation between the studies (only 8 of the 28 had external validation). So further work on independent datasets is still required. My target is to achieve a sensitivity of at least 90% and an AUC of at least 0.95. These are targets that we're aiming for, not results that we've already achieved. I will also look at the errors for individual findings as a single overall score could mask poor performance on some of the less common conditions. If the targets aren't met we will investigate whether the errors are due to negation, uncertain labels or limitations in the dataset.

Conclusion

Task focused ClinicalBert will be the main model, while TF-IDF with logistic regression and BioBert will be used for comparision. The models will be evaluated using a consistent experimental setup and appropriate performance measures. The aim is not to assume that the most advanced model will automatically perform best, but to compare the approaches fairly and understand which one extracts clinical information more reliably. The base paper provides the research direction, while my project will test the proposed approach using a focused public dataset. The final conclusions will depend on the experimental results, and all performance values will be reported honestly.